



Stable bubble oscillations beyond Blake's critical threshold



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ABSTRACT

The equilibrium radius of a single spherical bubble containing both non-condensable gas and vapor is determined by the mechanical balance at the bubble interface. This expression highlights the fact that decreasing the ambient pressure below the so called Blake's critical threshold, the bubble has no equilibrium state at all. In the last decade many authors have tried to find evidence for the existence of stable bubble oscillation under harmonic forcing in this regime, that is, they have tried to stabilize the bubble motion applying ultrasonic radiation on the bubble. The available numerical results provide only partial proof for the existence as they are usually based on linearized or weakly nonlinear (higher order approximation) bubble models. Here, based on numerical techniques of the modern nonlinear and bifurcation theory, the existence of stable bubble motion has been proven without any restrictions in nonlinearities. Although the model, applied in this paper, is the rather simple Rayleigh–Plesset equation, the presented technique can be extended to more complex bubble models easily.

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1. Introduction

The equilibrium radius of a gas- and/or vapor-filled spherical bubble is characterized by the well-known static mechanical equilibrium at the bubble interface. For a given mass of gas m_G and ambient temperature T_∞ it is written as

$$0 = p_V - P_\infty + p_{go} \left(\frac{R_o}{R_E} \right)^{3n} - \frac{2\sigma}{R_E}, \quad (1)$$

where R_E is the equilibrium bubble radius, $\sigma(T_\infty)$ is the surface tension, $p_V(T_\infty)$ is the vapor pressure and P_∞ is the static ambient pressure far away from the bubble. The reference pressure and radius and the exponent of the polytropic state of change are p_{go} , R_o and n , respectively. The reference quantities determine the mass of gas within the bubble:

$$m_G = \frac{4p_{go}R_o^3\pi}{3\Re T_\infty}, \quad (2)$$

where $\Re$ is the specific gas constant. Observe that Eq. (2) corresponds to equilibrium conditions, thus, the constant ambient temperature T_∞ does not imply isothermal gas behavior.

Typical equilibrium bubble radius curves are presented in Fig. 1 for water at $T_\infty = 37^\circ\text{C}$. The stable R_E^s and unstable R_E^u radii are marked by the solid and dashed lines, respectively. If the bubble interior contains non-condensable gas beside the vapor, that is, $m_G > 0$ than a turning point appears in the equilibrium radius

curve called Blake's critical threshold, which first identified by Blake [6] and later on Neppiras and Noltingk [23], see the back dot in Fig. 1. Throughout this paper the equilibrium radius at this critical point will be referred to as critical radius R_C . It is important to emphasize that beyond Blake's threshold there is no equilibrium state of the bubble. In the absence of foreign gas ($m_G = 0$), however, no stable equilibrium exists at all, resulting in an infinite growth or a total collapse of the vapor bubble for all values of the ambient pressure. The above-described graphical representation was used by Daily and Johnson [8] and a thorough discussion of the equilibrium state of a bubble can also be found in Brennen [7].

Although the above description excludes the existence of stable vapor bubbles, the experiment of Marston and Greene [20] reported the observation of stable microscopic vapor bubbles under a harmonically varying pressure field, discovering the fact that ultrasonic radiation may stabilize the bubble motion. The authors also concluded that the stability may be influenced by the previously found second (evaporation–condensation) resonance. Beside the so-called Minnaert primary resonant frequency [21] this second resonance frequency was first reported by Neppiras and Finch [22] in vapor bubbles for a given equilibrium radius. Their observation was supported by other authors, see Wang [29] or Wang [30]. The discovery of this phenomenon had triggered an increasing attention of the researchers on the subject. Several versions of analytical approximations for this second resonance were proposed, with different degrees of complexity, by Patel et al. [27], Marston [19], Hsieh [16], Akulichev [1], Alekseev [2] and many others. Since these studies are based on first or second order approximations, their results are limited only to small amplitude oscillations.

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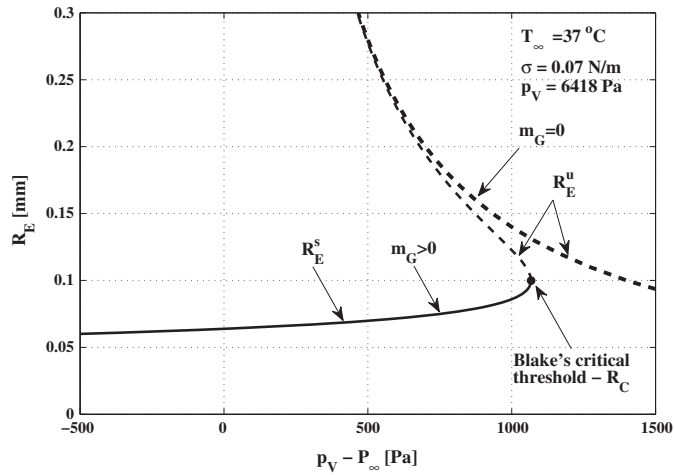


Fig. 1. Typical equilibrium bubble radius curves for water at $T_\infty = 37^\circ\text{C}$. The solid and dashed lines are the stable and unstable equilibrium radius curves, respectively. The thick dashed curve corresponds to pure vapor bubbles, whereas in the case of the thin curve the bubble interior contains both vapor and foreign non-condensable gas. The black dot denotes the so-called Blake's critical threshold.

These results encouraged Hao and Prosperetti [12] to solve a relatively complex bubble model numerically, without any linearization or other approximations, to verify the predictions of the linear theories and try to achieve stable bubble oscillations under harmonic forcing. Unfortunately, they could not find any stable oscillation even after thousands of cycles. A thorough analysis of Gumerov [10] provided a detailed explanation on the stability of vapor bubbles and revealed that in the parameter range applied by Hao and Prosperetti [12] there is no stable bubble oscillation.

The valuable results of Gumerov [10] proved that may be possible to stabilize a vapor bubble with harmonic forcing even if there is no any stable equilibrium or any equilibrium solution of the unexcited system. However, the question of this stabilization mechanism for large amplitude (nonlinear) oscillations is still open due to the applied third order approximation. The exponentially increasing computational resources, the rapidly developing and spreading nonlinear theory and its numerical techniques could help to overcome the problem. Still, the majority of the corresponding papers of the topic use less sophisticated techniques, such as the so-called brute force method, hence often miss the chance to find stable solutions, see again Hao and Prosperetti [12]. Even if some authors make use of the benefit of the advanced numerics the applied parameter values are simply out of the range of interest, see Parlitz et al. [25], Simon et al. [28], Behnia et al. [3–5]. In these cases the bubble contains non-condensable gas ($m_G > 0$) and the ambient pressure is so high that stable bubble oscillations always exist (far below Blake's critical threshold). One of the most recent review paper summarizing the achievements of nonlinear dynamics on bubble oscillations is written by Lauterborn and Kurz [18].

The main aim of this study is to prove that periodic forcing can stabilize the bubble motion even if there is no stable bubble equilibrium or any equilibrium radius of the unexcited system. Although the employed bubble model is rather simple, the well-known Rayleigh–Plesset equation, there is no restriction (linearization or other reduced order modelling) on our numerical technique. This means that the results are valid also for large amplitude oscillations, of course within the accuracy of the physical modelling. The surrounding of the bubble is pure water and the bubble content is a mixture of water vapor and non-condensable gas ($m_G > 0$). Therefore, the region where stable equilibrium does not exist lies beyond Blake's critical threshold, see Fig. 1. The procedure described here to find stable solutions can be easily

applied to more complex models and for pure vapor bubbles ($m_G = 0$).

2. Mathematical model

The model describing the evolution of the bubble radius in time is rather simple. Integrating the one dimensional Navier–Stokes equation (incompressible liquid) from the bubble radius $R(t)$ to infinity making use of the equation of continuity, one can obtain the equation

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho_L}(p_L - p_\infty) \quad (3)$$

known as the Rayleigh–Plesset equation which is a second order ordinary differential equation. Here, ρ_L is the density of the liquid, p_L is the pressure at the bubble wall in the liquid domain and p_∞ is the pressure far away from the bubble consisting of static and periodic components written as

$$p_\infty(t) = P_\infty + p_A \sin(\omega t), \quad (4)$$

where P_∞ is the ambient pressure, p_A and ω is the pressure amplitude and angular frequency of the excitation, respectively. The bubble interior contains both vapor and non-condensable gas assuming ideal gas behavior. The pressure of this ideal mixture is composed by the sum of the partial pressure of vapor p_V and the gas p_G . The connection between the pressure inside and outside the bubble at the bubble interface is expressed by the mechanical balance

$$p_G + p_V = p_L + \frac{2\sigma}{R} + 4\mu_L \frac{\dot{R}}{R}, \quad (5)$$

where σ is the surface tension and μ_L is the liquid dynamic viscosity. The vapor pressure inside the bubble is constant but its value depends on the ambient temperature T_∞ while the gas content obeys a simple polytropic relationship

$$p_G = p_{g0} \left(\frac{R_0}{R} \right)^{3n} \quad (6)$$

with a polytropic exponent n . Although n can vary between 1 and 1.4 (isothermal and adiabatic behavior), this study follows the concept of Lauterborn and Kurz [18] and $n = 1.4$ was applied. The reference pressure p_{g0} and radius R_0 determine the mass of gas inside the bubble, see Eq. (2). The material properties of the liquid domain (water) and the water vapor inside the bubble were computed by means of the Haar–Gallagher–Kell equation of state (see [11]) at T_∞ and P_∞ , whose values are discussed in the next subsection.

2.1. Parameters and their transformations

Before presenting any specific computations, let us summarize the parameters needed to specify in the model. This paper follows the idea dealing with the large number of parameters, first proposed by Hegedüs et al. [13], based on the fact that the material properties of a pure substance depend on the pressure P_∞ and temperature T_∞ . Regarding these two quantities as independent parameters, all the material properties can be determined, hence, the number of parameters can be significantly reduced, especially in complex bubble models. Beside the ambient properties a further parameter for the description of the bubble size is still needed. This can be, for instance, the mass of gas within the bubble (m_G), the equilibrium bubble radius (R_E) or the critical radius (R_C , see the investigation below). Finally, in the presence of harmonic forcing the pressure amplitude p_A and angular frequency ω are also required.

First, let us discuss the description of the bubble size. The mass of gas m_G is not an illustrative measure. The equilibrium radius R_E

varies with the ambient pressure and vanishes as the tension $p_V - P_\infty$ pass beyond Blake's critical threshold. Thus, the proper choice is the critical radius R_C of the unexcited system, explained below. At the location of the critical point (turning point) the derivative of the tension with respect to the equilibrium radius is zero given by the expression

$$\left. \frac{d(p_V - P_\infty)}{dR_E} \right|_{R_E=R_C} = -\frac{2\sigma}{R_C^2} + 3np_{go} \frac{R_0^{3n}}{R_C^{3n+1}} = 0, \quad (7)$$

compare also with Eq. (1). If the reference radius R_0 is chosen to be the critical radius R_C the gas reference pressure can be expressed from Eq. (7) as

$$p_{go} = \frac{2\sigma}{3nR_C}. \quad (8)$$

By substituting $R_0 = R_C$ and expression (8) into Eq. (2) it becomes clear that the critical radius alone determines the mass of gas inside the bubble

$$m_G = \frac{8\sigma R_C^2 \pi}{9nRT_\infty}, \quad (9)$$

and therefore can characterize the size of the bubble for known material and at a given ambient temperature T_∞ . During the computations the value of the constant critical radius was chosen to be $R_C = 0.1$ mm.

The ambient pressure at the critical point (critical ambient pressure P_C) can also be expressed with the known critical radius as

$$P_C = p_V - \frac{2\sigma}{R_C} \frac{3n-1}{3n}, \quad (10)$$

with the substitution $R_0 = R_C$ and expression (8) into Eq. (1). Introducing a dimensionless relative pressure P_R the transformation

$$P_\infty = p_V + (P_C - p_V)P_R \quad (11)$$

maps the region of ambient pressure $P_\infty \in (p_V, P_C)$ into $P_R \in (0, 1)$. In other words, $P_R = 0$ means zero tension ($p_V - P_\infty = 0$) and in case of $P_R = 1$ the tension is at the critical threshold. If the relative pressure is $P_R < 0$, $0 < P_R < 1$ or $P_R > 1$, then the number of equilibrium points in the unexcited system is 1, 2 or 0, compare also with Fig. 1. The present study attempts to answer the most important questions, namely, is it possible to stabilize the bubble motion with harmonic forcing as the relative pressure exceeds 1 (pass beyond Blake's critical threshold) or not? Observe that varying relative pressure means varying ambient pressure. The other ambient property, the ambient temperature was kept constant $T_\infty = 37^\circ\text{C}$ during the computations.

The remaining two parameters are related to the excitation itself, namely, the pressure amplitude p_A and the angular frequency ω , see Eq. (4). As the angular frequency can vary between several orders of magnitude, its normalization with a suitable reference quantity is reasonable. The peak frequency corresponding to a stable equilibrium radius is

$$\omega_p = \sqrt{\frac{3n(P_\infty - p_V)}{\rho_L R_E^2} + \frac{2(3n-1)\sigma}{\rho_L R_E^3} - \frac{8\mu_L^2}{\rho_L^2 R_E^4}}, \quad (12)$$

see for instance Brennen [7]. Due to the absence of any equilibrium radius R_E at very low ambient pressures P_∞ , it is replaced by the critical radius R_C . Furthermore, to avoid complex frequencies because of the possible unusual parameter values leading to negative sign under the root, Eq. (12) was simplified to the expression

$$\omega_{ref} = \sqrt{\frac{4\sigma}{\rho_L R_C^3}} \quad (13)$$

leading to a new variable. Now, the applied relative frequency during the computations is defined as

$$\omega_R = \frac{\omega}{\omega_{ref}}. \quad (14)$$

The pressure amplitude p_A and relative frequency ω_R are the main two control parameters of the present investigations.

2.2. Dimensionless equation system

Introducing the dimensionless bubble radius $y_1 = R/R_C$, the dimensionless time $\tau = t/(2\pi/\omega)$ and defining a dimensionless bubble wall velocity as $y_2 = y_1'$, where ' stands for the derivative with respect to τ , the governing second order nonlinear differential equation can be rewritten as a system of first order dimensionless equations:

$$y_1' = y_2, \quad (15)$$

$$y_2' = \frac{p_L - p_\infty}{p_{ref}} \frac{1}{y_1} - \frac{3}{2} \frac{y_2^2}{y_1}, \quad (16)$$

where the reference pressure is

$$p_{ref} = \rho R_C^2 \left(\frac{\omega}{2\pi} \right)^2. \quad (17)$$

The pressure far away from the bubble is

$$p_\infty(t) = P_\infty + p_A \sin(2\pi\tau), \quad (18)$$

the form of the expression of the mechanical balance at the bubble interface becomes

$$p_G + p_V = p_L + \frac{2\sigma}{R_C} \frac{1}{y_1} + \frac{4\mu_L \omega}{2\pi} \frac{y_2}{y_1}, \quad (19)$$

and finally the polytropic relationship of the gas content is

$$p_G = \frac{2\sigma}{3nR_C} \left(\frac{1}{y_1} \right)^{3n}. \quad (20)$$

3. Response to harmonic forcing

3.1. General overview of the numerical techniques

The objective of the present paper is to find stable solutions of the excited system, Eqs. (16)–(20), in the ambient pressure regime where stable equilibrium radius does not exist in the absence of the harmonic forcing. In this case the ambient pressure is so low that the tension exceeds the so called Blake's critical threshold, see again Fig. 1. Because of the strong nonlinearity of the system, the number of the possible stable structures is high. Various periodic, with distinct periods, and chaotic solutions can coexist even at the same parameter values. Therefore, this investigation focuses only on the existence of the stable period 1 solutions. Period 1 solution means that the trajectory returns to its starting point after one cycle of the excitation.

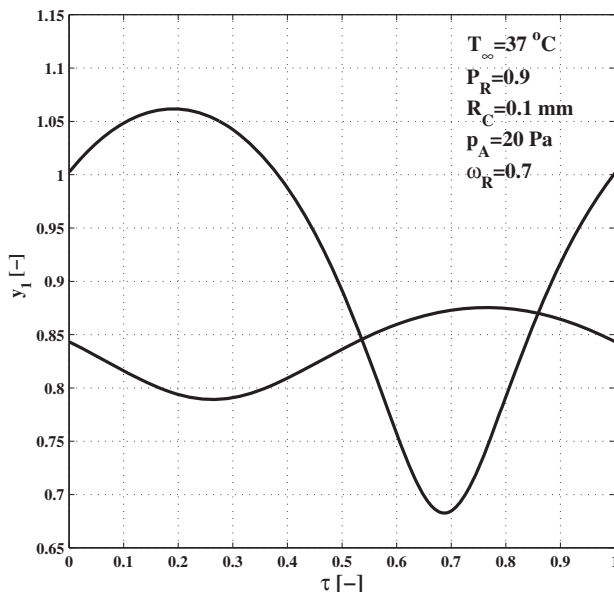
The simplest and still widely used way to find stable periodic solutions is to take an initial value problem (IVP) solver and start a simulation from a properly chosen initial condition. After many, sometimes even hundreds, of cycles the solution usually converges to a stable structure (attractor) if there is any at the given parameter values. The main advantage of this method is its simplicity, but the unstable solutions cannot be detected even if one integrates backward in time (usually there are saddle-type unstable structures). The rapidly developing nonlinear bifurcation theories and their numerical techniques opened the way to find the unstable solutions. These are important, for instance, in the determination of the domain of attraction of the stable solutions, thus, they

give a better understanding in the global dynamics of the system. The idea is to apply a boundary value problem (BVP) solver with periodic boundary conditions and compute the desired periodic solution directly. Due to the insensitivity of the method to stability, its combination with the very efficient pseudo-arclength continuation technique complete bifurcation curves can be traced with respect to a single control parameter (codim 1 bifurcation curve). Moreover, the bifurcation points, the locations where the stability changes, can also be detected along the curve (for example saddle-node or period doubling bifurcations). The main strength of this method is the possibility to compute the detected bifurcation points also in a two dimensional parameter space by choosing a secondary control parameter (codim 2 bifurcation curve). A severe drawback, however, is the necessity of a very good initial solution for the convergence. Therefore, the best practice is the usage of the IVP and BVP solvers simultaneously.

In this paper the IVP solver was used to obtain a global picture about the attractors by computing several bifurcation curves, choosing the pressure amplitude p_A as the main control parameter at a relative pressure $P_R = 0.95$ and at several values of the relative frequency ω_R . All the other parameters were kept constant. These maps can reveal the possibly coexisting stable period 1 solutions. Initiating the BVP solver with the previously obtained IVP solutions, complete bifurcation curves of period 1 solution can be determined including the unstable solutions and the bifurcation points. Now, it is possible to compute the curves of the detected bifurcation points in the $p_A - \omega_R$ parameter space (phase diagram) by choosing the relative frequency ω_R as the secondary control parameter. These curves enclose the regions where stable period 1 solutions exist. Finally, gradually increasing the relative pressure P_R and exploiting the advantage of the combination of the BVP solver and the pseudo-arclength continuation technique, the domains of the stable solutions in the $p_A - \omega_R$ plane are revealed up to the relative pressure $P_R = 1.01$. Keep in mind again that $P_R > 1$ means exceeding Blake's critical threshold.

3.2. Examples of period 1 solutions

Two examples for the desired period 1 attractors are shown in Fig. 2 computed with an IVP solver in MATLAB software environment at the given parameter values. These were as follows:



ambient temperature $T_\infty = 37^\circ\text{C}$, relative pressure $P_R = 0.9$, critical radius $R_C = 0.1$ mm, pressure amplitude $p_A = 20$ Pa, relative frequency $\omega_R = 0.7$. With these fixed parameter values the presented solutions coexist; each having its own domain of attraction, and thus, the applied initial condition determines the orbit to which its corresponding transient solution converges.

Observe that the period of the excitation is unity ($\tau_0 = 1$), see Eq. (18), meaning that the period 1 solutions return to itself after every integer value of the dimensionless time τ . This behavior can be clearly seen in the left hand side of Fig. 2 which represents the radius vs time curves of the attractors.

Naturally, the trajectories of the periodic orbits are closed curves in the dimensionless phase space ($y_1 - y_2$ plane), as displayed in the right hand side of Fig. 2. The black dots are the points of the so called Poincaré map obtained by sampling the continuous solution at time instants $\tau = kT_0$, where $k \in \mathbb{N}$. Initiating the system from an arbitrary point y_0 , its corresponding iterated value is denoted by $P(y_0)$. If a trajectory returns exactly to the starting value after N iterations ($P^N(y_0) = y_0$), then the solution is a periodic orbit whose period is exactly $T_p = NT_0$ called period N solution. Specifically in our period 1 cases the Poincaré maps return to itself immediately after the first period, that is, $y_0 = P(y_0) = P^2(y_0) = \dots$, see for example the internal orbit in the right hand side of Fig. 2.

3.3. IVP solutions to map the coexisting attractors

In order to reveal the above described coexistence and to obtain a global picture about the period 1 attractors in the pressure amplitude-relative frequency ($p_A - \omega_R$) parameter space, various IVP simulations were performed at several pressure amplitudes and relative frequencies. The solver was a standard 5th order Runge-Kutta scheme with 4th order embedded error estimation. During a complete simulation the control parameter was the pressure amplitude while the relative frequency was kept constant. The pressure amplitude was increased from 10 Pa to 10,000 Pa with an increment of 1 Pa. At each pressure amplitude 10 simulations were carried out with random initial values to reveal the coexisting attractors. After the convergence of each solution 32 points from the Poincaré plane were recorded.

In Fig. 3 two pressure amplitude response curves (bifurcation diagrams) are presented at relative frequencies 0.7 and 5.0

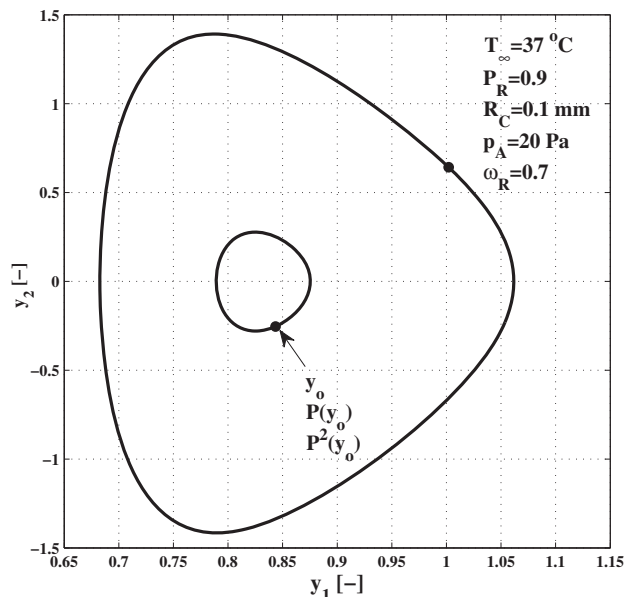


Fig. 2. Two examples of period 1 attractors. The continuous solutions return to themselves after one period of the excitation $\tau_0 = 1$ (left). The trajectories in the phase space (right) are closed curves. The black dots represent points in the Poincaré maps of the solutions, returning to themselves after every period $y_0 = P(y_0) = P^2(y_0) = \dots$

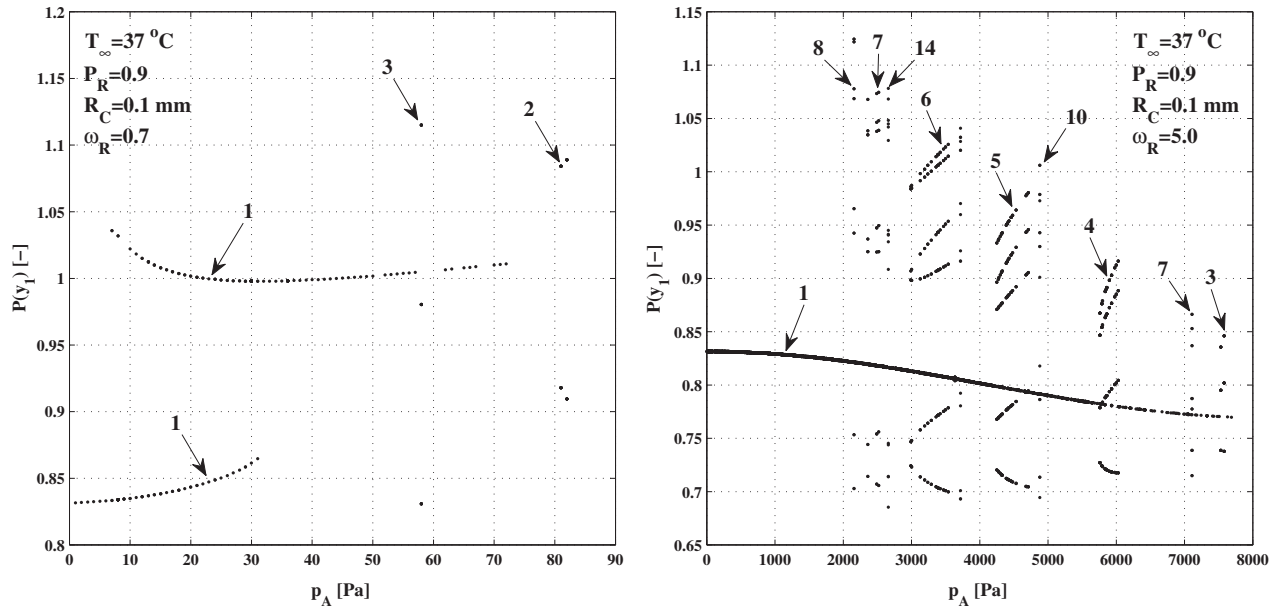


Fig. 3. Pressure amplitude response curves of IVP computations at different relative frequencies ω_R with the pressure amplitude p_A as control parameter. The numbers denote the periods of the found attractors.

computed with the IVP solver. The periods of the different kinds of attractors are denoted by the arabic numbers. Although the maximum pressure amplitude was 10,000 Pa the limits of the bifurcation diagrams are usually much less due to the applied positive tension on the system ($P_R > 0$). In this case stable bubble motion is not guaranteed and the bubble can grow infinitely, as discussed in Section 1. The chance to find a stable solution depends on its domain of attraction. From Fig. 3 it is clear that an increasing pressure amplitude decreases this domain; even if stable structures exist, they are hard to find with a randomly initiated IVP solver.

At low relative frequency ($\omega_R = 0.7$) the system is rather simple; the only noteworthy property is a hysteresis, shown in the left hand side of Fig. 3. In the pressure amplitude range of this structure two distinct period 1 attractors coexist. Observe that in Fig. 2 a pair of these stable solutions are presented at pressure amplitude $p_A = 20$ Pa. On the other hand, at a high relative frequency, such as $\omega_R = 5.0$ the system is very feature-rich from a dynamical point of view, as seen in the right hand side of Fig. 3. In this case various kinds of periodic attractors in a wide range of the pressure amplitude were found, up to period 14. During the present investigation similar bifurcation diagrams, as those in Fig. 3 were computed at various relative frequencies ranging from 0.1 to 5.0 with 0.1 increment.

The above-described IVP solutions can serve as accurate initial solutions for the more sophisticated BVP technique as it needs a very good initial guess in order to converge properly. Global exploration of the possible attractors with an IVP solver is always advisable, because the dynamics of the system can be so complicated that one can miss some features of the system.

3.4. Pressure amplitude response curves computed with the BVP solver

The IVP solver gave a rough picture about the period 1 and other kinds of attractors in the $p_A - \omega_R$ parameter space. The BVP method together with the pseudo-arclength continuation technique, however, can map all the period 1 solutions, both the stable and unstable ones, including the detection of the bifurcation points where the change of stability takes place. These sophisticated pressure amplitude response curves were obtained by initiating the

BVP solver from the results of the IVP computations at the previously mentioned values of the relative frequency.

During the simulations the AUTO continuation software was employed, see Doedel et al. [9]. Because AUTO can handle only autonomous systems (free of explicit time dependence), Eq. (16) has to be extended with two additional decoupled ODEs as follows:

$$y'_1 = y_2, \quad (21)$$

$$y'_2 = \frac{p_L - p_\infty}{p_{ref}} \frac{1}{y_1} - \frac{3}{2} \frac{y_2^2}{y_1}, \quad (22)$$

$$y'_3 = y_3 + 2\pi y_4 - y_3(y_3^2 + y_4^2), \quad (23)$$

$$y'_4 = -2\pi y_3 + y_4 - y_4(y_3^2 + y_4^2), \quad (24)$$

where the periodic solution of the variable y_4 is exactly $-\sin(2\pi\tau)$ and thus the pressure far away from the bubble becomes

$$p_\infty(t) = P_\infty - p_A y_4. \quad (25)$$

This description has a severe drawback, that is, AUTO can handle a periodic solution as a whole object regardless of their periods. Therefore, the illustrative representation in the Poincaré map has been lost and one can obtain only the maximum value of the dimensionless bubble radius of each kind of solution.

Fig. 4 shows some pressure amplitude response curves at different relative frequencies. The solid and dashed lines are the stable and unstable solutions, respectively. Two types of bifurcations have been experienced. The saddle-node also known as fold (FL) bifurcation marked by dots, in which a stable (node-like solution) and an unstable (saddle-like solution) collide. Observe that two FL points form a single hysteresis, see for instance the enlargement of Fig. 4C. The second type is the period doubling (PD) bifurcation denoted by the crosses, where the originally stable solution loses its stability and a stable solution is bifurcated with doubled period.

At the high relative frequency ($\omega_R = 5.0$) the structure of the period 1 solutions is rather simple, see Fig. 4A. From the stable equilibrium point R_E^s ($p_A = 0$) a stable periodic orbit is bifurcated as soon as a small, but definitely non-zero pressure amplitude is applied on the system (lower curve). This solution loses its stability through a PD bifurcation at $p_A = 9256$ Pa and remains unstable with increasing pressure amplitude. Observe that at relative pressures $0 < P_R < 1$ the unexcited system has also an unstable

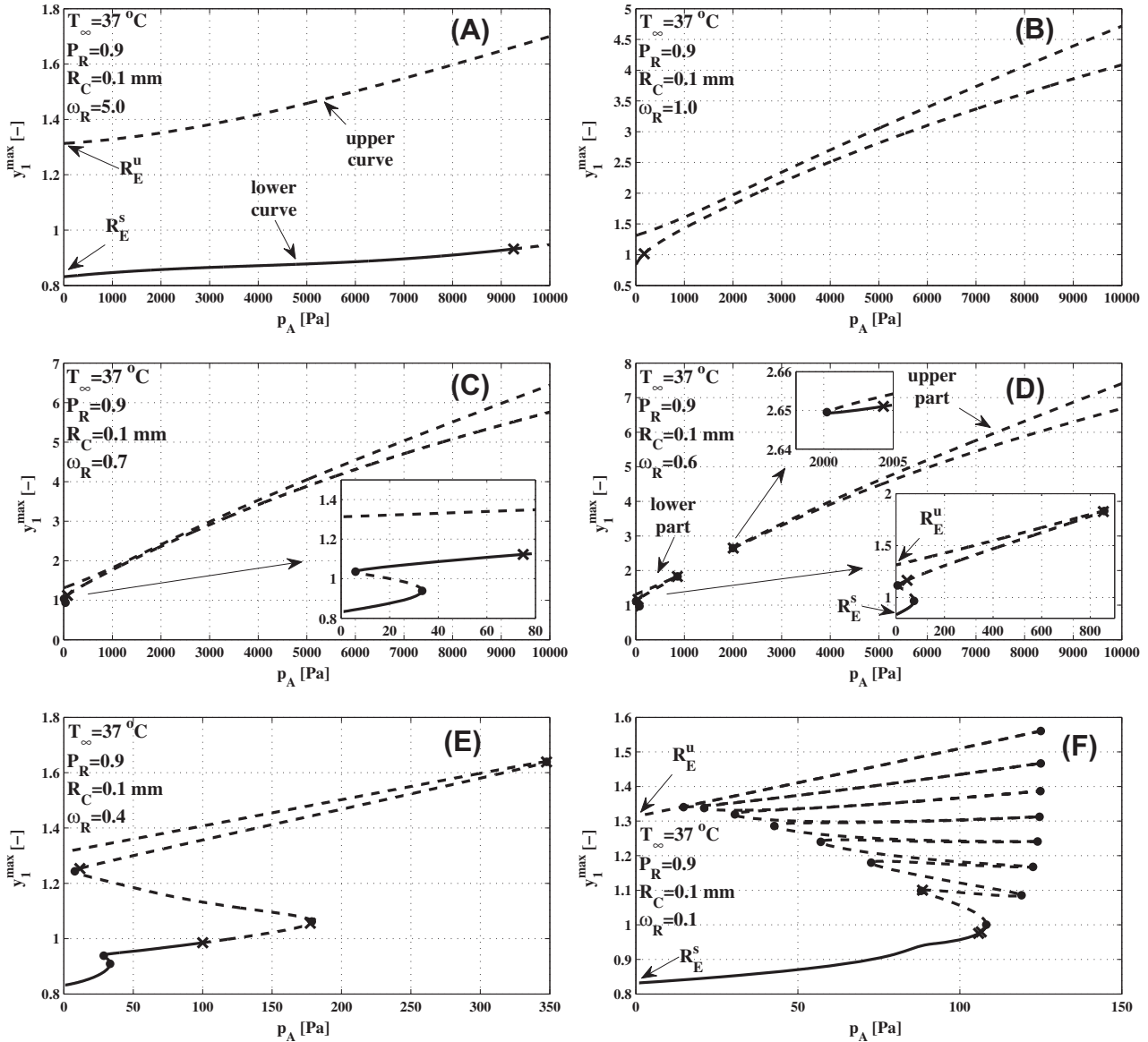


Fig. 4. Period 1 bifurcation curves at different relative frequencies ω_R . The control parameter is the pressure amplitude p_A . The solid and dashed lines represent the stable and unstable solutions, respectively. The period doubling and fold bifurcation points are denoted by crosses and dots, respectively. For the more detailed explanation of the figures see the text.

equilibrium point R_E^u , see the region of the tension $p_V - P_\infty$ between zero and Blake's critical threshold in Fig. 1. Due to the same mechanism as in the case of R_E^s a secondary unstable period 1 orbit bifurcates from R_E^u under harmonic forcing, denoted by the dashed line in Fig. 4A (upper curve). This solution remains unstable for all values of the control parameter p_A , therefore, it is impossible to find it with the IVP solver. Here a rough initial guess for the BVP computations could be obtained by simulating with the IVP solver only one period initiating from the unstable equilibrium point at $p_A = 10$ Pa.

As the relative frequency ω_R decreases the steepness of the curves increase, the PD bifurcation takes place at a smaller and smaller value of the pressure amplitude and the two curves get closer to each other, see Fig. 4B. At $\omega_R = 0.7$ the first hysteresis appear in the lower curve formed by two FL bifurcations, see Fig. 4C and its magnification. Compare the results in the enlargement with Fig. 3 left and also with the simulations presented in Fig. 2 at $p_A = 20$ Pa. The upper and lower branches are getting even closer with decreasing ω_R and finally they collide somewhere between the relative frequency values 0.7 and 0.6 separating the structure into a

upper ($p_A > 2000$ Pa) and a lower ($p_A < 1000$ Pa) part, see Fig. 4D. In the following discussions it becomes clear that this separation plays an important role in the existence of stable solutions at $P_R > 1$ (beyond Blake's critical threshold). The upper part contains only an FL and a PD bifurcation point. Between them there is a short section of stable solutions, see the upper magnification of Fig. 4D. This property of the upper part holds for all the lower relative frequencies. The lower part contains still the sole hysteresis and now as a single curve it connects the stable and the unstable equilibrium radii, shown in the lower magnification of Fig. 4D.

In the rest of the subfigures (Fig. 4E and F) only the lower part of the period 1 structures are presented at relative frequencies $\omega_R = 0.4$ and $\omega_R = 0.1$ (the upper part remains qualitatively the same). It is clearly seen, that hystereses (FL pairs) and PD windows gradually appear one after another as the relative frequency decreases. At $\omega_R = 0.1$ there are 7 hystereses and PD windows altogether, see Fig. 4F. Observe that the PD bifurcation points after the first hysteresis (starting from R_E^s) are so close to the FL points that all the structures after the first FL bifurcation are practically unstable.

The same technique was used (combination of the IVP and BVP solvers) by Hegedüs et al. [14], to obtain parametric maps for the stable period 1, 2 and 3 solutions at higher ambient pressure, and by Hegedüs and Kullmann [15] for the determination of the domains of attraction of coexisting stable solutions near the first subharmonic resonance. Both study were applied the same Rayleigh–Plesset bubble model.

3.5. Phase diagram at $P_R = 0.9$

It is clear from the computed pressure amplitude response curves that the detected bifurcations determine the number of the period 1 attractors. For instance, at $\omega_R = 0.7$ (Fig. 4C) there is only one stable solution below the first FL point, there are two attractors in the range of the hysteresis (between the first and second FL points), between the second FL and the PD points there is again only one stable solution and finally, over the PD point there is no stable period 1 solution at all. With the AUTO continuation software these bifurcation points can be traced in the $p_A - \omega_R$ plane by choosing ω_R as a secondary continuation parameter. The resulting PD and FL curves enclose the domain of the stable period 1 solutions shown in Fig. 5 by the gray regions, where the FL and PD bifurcations are the solid and dashed curves, respectively. By far the largest area is located underneath the period doubling curve marked by PD_0 . This kind of diagrams will be referred to as phase diagrams in the rest of the paper, following the terminology of the literature.

The low frequency and amplitude parameter range (enlargement-1 in Fig. 5) is interesting only from the dynamical point of view, because, the size of the domain of the attractors here is negligible compared to the extent of the whole investigated parameter space. Observe that the detected PD and FL points of the pressure amplitude response curve at $\omega_R = 0.7$ in Fig. 4C are marked by the cross and the two dots, respectively, in enlargement-1. The sole PD point corresponds to the curve PD_0 , while the two FL points, which form a hysteresis, belong to the same bifurcation curve

denoted by FL_0 . The upper and lower branches of the curve FL_0 approach with increasing relative frequency and collide approximately at $\omega_R = 0.8201$ with the consequence of the disappearing hysteresis. The point of the collision is called cusp bifurcation, in the literature of bubble dynamics it is often referred to as resonance horn. Up to the relative frequency $\omega_R = 0.1$ there are 7 cusp bifurcations altogether, which means that there must be 7 hysteresees in the corresponding pressure amplitude response curve, see also Fig. 4F. The cusp point having the highest frequency value is usually called the main resonance, while the subsequent cusp bifurcations are the harmonic resonances.

The other interesting feature of the system is related to the bifurcation curves FL_1 and PD_1 , see enlargement-2 in Fig. 5. These are responsible for the collision of the upper and lower pressure amplitude response curves near $\omega_R = 0.6$ shown in Fig. 4C and D. First, a PD window appears in the lower curve at $\omega_R = 0.6449$ forming a small segment of stable solution; observe also the developing thin gray area between the curves PD_1 and FL_1 . Lowering further the frequency the collision takes place at $\omega_R = 0.6439$ through the appearance of two FL points in the stable segment forming the FL_1 curve. This process resulted in the separation of the whole pressure amplitude response bifurcation diagram into an upper and a lower part, see again Fig. 4D and the discussion of the previous section. It has already been mentioned before that this separation (PD_1, FL_1 curves) plays an important role in the formation of stable solution at $P_R > 1$ (beyond Blake's critical threshold) explained in more detail in the next subsection.

3.6. The evolution of the phase diagram with increasing relative pressure

Before continuing with the discussion let us summarize the so far acquired results. With a BVP solver together with the pseudo-arc-length continuation technique pressure amplitude response curves were computed including the detection of the bifurcation points at several relative frequencies ω_R and at the relative

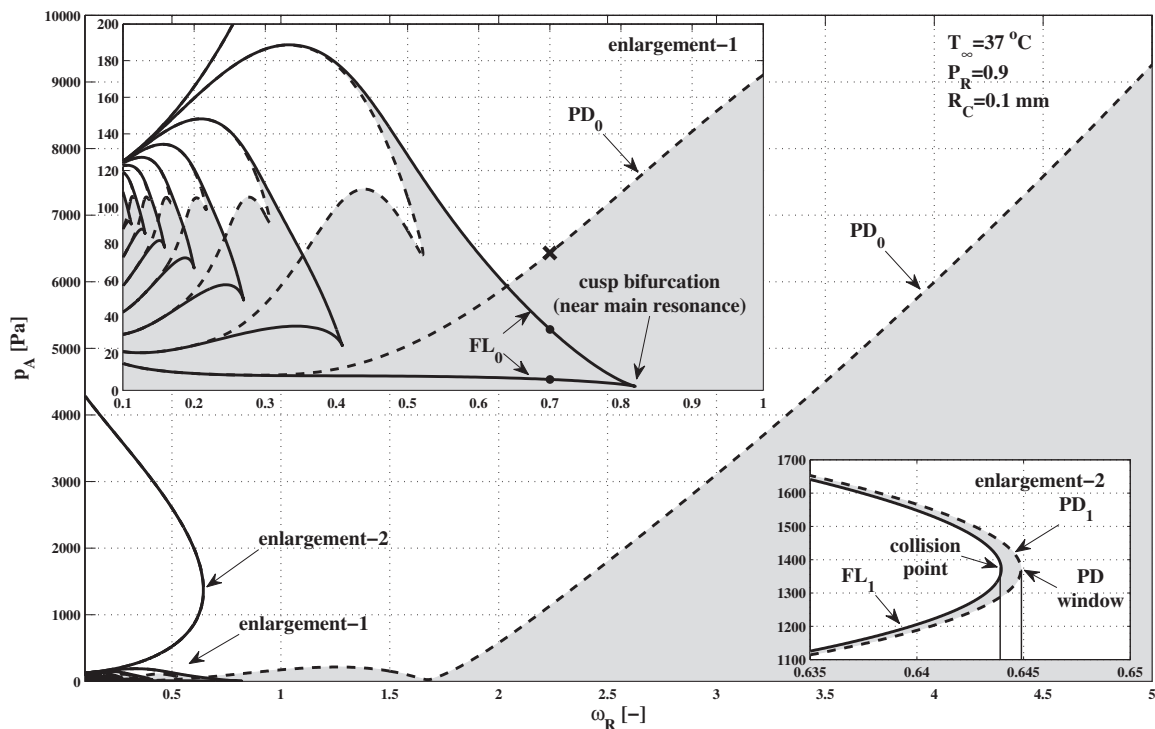


Fig. 5. Phase diagram of the fold (solid curves) and the period doubling (dashed curves) bifurcation points in the $p_A - \omega_R$ parameter plane corresponding to the period 1 solutions. The existence of period 1 attractors is indicated by the gray domain.

pressure $P_R = 0.9$. A single curve may contain both stable and unstable solutions and various numbers of folds and hystereses. The detected period doubling (PD) and saddle-node (FL) bifurcations were traced in the pressure amplitude – relative frequency ($p_A - \omega_R$) parameter space. These curves enclose the domain of the period 1 attractors. The whole computational procedure was successful due to the fact that the unexcited system ($p_A = 0$) at $P_R = 0.9$ has equilibrium points from which periodic solutions bifurcated as the pressure amplitude gradually increased, demonstrated by the lower and upper bifurcation curves in Fig. 4.

Now, there is only one remaining question, namely, is there any part of the stable domain which survives as the relative pressure exceeds 1 or not? The question can be easily answered by repeating the above-described procedure and compute several phase diagrams for different values P_R . The main virtue of the BVP solver is that one needs only one period 1 solution at a specific P_R , whether it is stable or unstable, and with the continuation procedure the whole phase diagram can be computed. By choosing the relative pressure P_R as the control parameter it was a minor problem to find some period 1 solution at $P_R = 0.92, 0.95, 0.99$ and at 1.01 by initiating the BVP solver from the results at $P_R = 0.9$. From these solutions the corresponding phase diagrams can be determined, shown in Fig. 6. The FL and PD bifurcation curves are again indicated by the solid and the dashed curves, respectively.

Although the domain of the period 1 attractors is not indicated with the filled areas in Fig. 6, the evolution of this domain is very clear. At $P_R = 0.92$ and at $P_R = 0.95$ (Fig. 6A and B) the phase

diagrams are qualitatively the same as presented in Fig. 5, only the number of the cusp bifurcation (hystereses) were reduced. The large domain of stable solutions still lies below the bifurcation curve marked by PD_0 . Increasing the relative pressure up to $P_R = 0.99$ a qualitative change can be found, see Fig. 6C. The bifurcation curve PD_0 and PD_1 collided and upper and lower period doubling bifurcation curves were formed marked by PD_{0-1}^I and PD_{0-1}^{II} , respectively. Observe that the number of the hystereses reduced significantly which means that the harmonic oscillations started to disappear. Finally, increasing the relative pressure to $P_R = 1.01$, all the harmonics and the curve PD_{0-1}^{II} vanish. The lower branch of the FL_1 curve jumped to the right hand side and a large domain of period 1 attractors formed between the curves FL_1 and PD_{0-1}^I . This large area proves that stable solutions exist even at relative pressures $P_R > 1$, that is, the bubble motion can be stabilized with harmonic forcing even if the unexcited system has no equilibrium solution at all. Moreover, these results are valid for large amplitude oscillations, since the advanced mathematical treatment requires no reduced order modelling, such as linearization or other higher order approximations.

Numerically, the existence of period 1 attractors beyond Blake's critical threshold has been proven. From the application point of view, however, the domain of attraction of these stable solutions could be an interesting question. If this domain is very small for a given attractor then it is hard to find it during an experiment, therefore, it is irrelevant in real situations. The exact domain of attraction is determined by the unstable manifolds of saddle-type

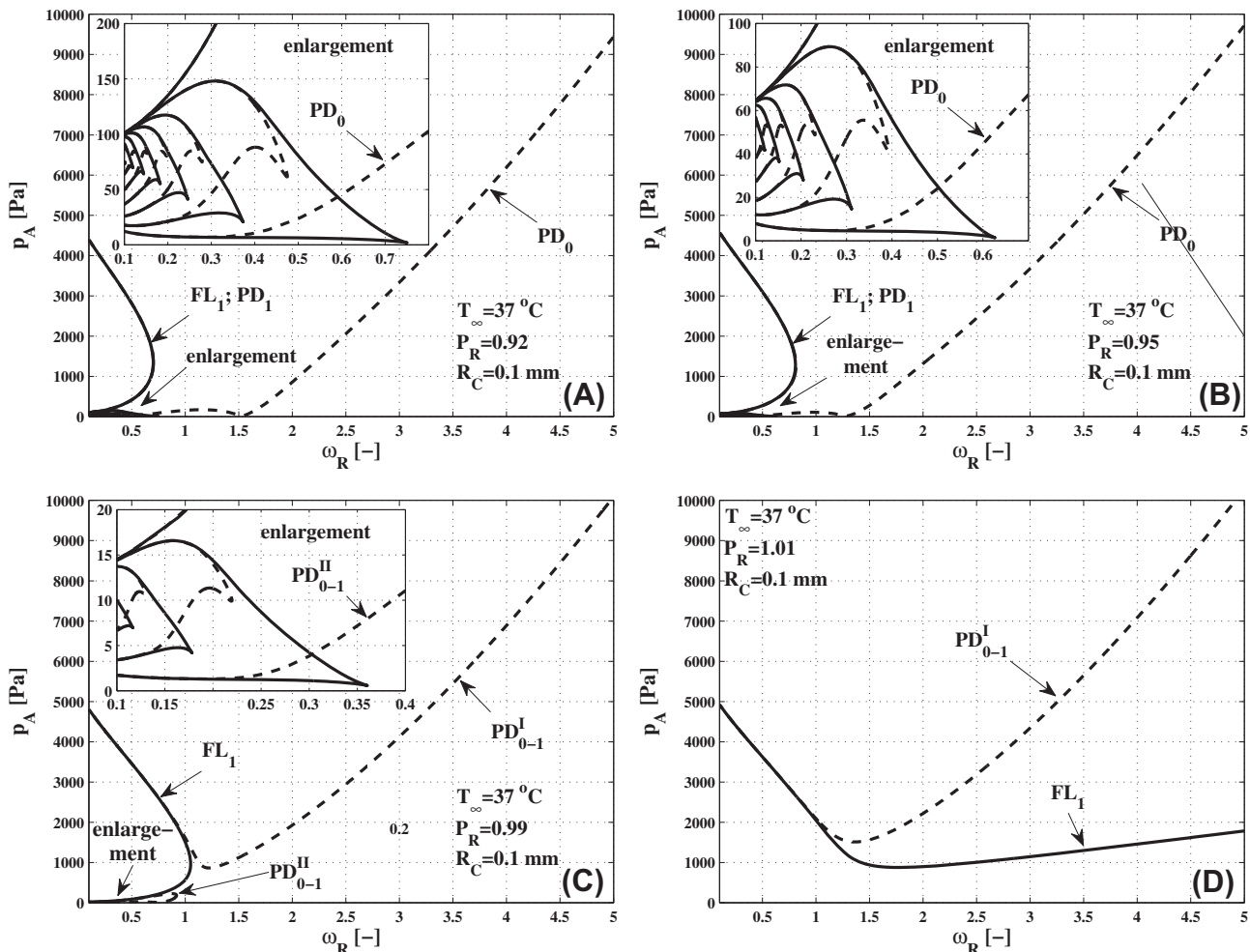


Fig. 6. Series of phase diagrams of the period 1 solutions at relative pressures $P_R = 0.92, 0.95, 0.99$ and 1.01. The fold FL and period doubling PD bifurcation curves are indicated by the solid and the dashed curves, respectively.

unstable solutions. This can be very resource demanding computations and therefore this problem is out of the scope of the present paper. Nevertheless, the large area of the period 1 attractors in Fig. 6D suggests that there may be solutions with significant domain of attraction at certain parameter values.

As the applied bubble model is rather simple, the classical Rayleigh–Plesset equation, the issue of the validity of the results should be discussed. In spite of the crude approximations of the model, it is reasonable to expect that for more complex models the phase diagrams would be qualitatively similar (topological equivalent). This is supported by the observation of the fact that several authors found very similar bifurcation structures in different oscillators, for instance, in the van der Pol oscillator Parlitz and Lauterborn [26], in the Toda oscillator Kurz and Lauterborn [17] or in the Duffing oscillator Parlitz [24]. Therefore, it is very likely that the above described stabilization mechanism is present in other more complex bubble models.

4. Conclusion

In spherical gas/vapor bubbles there are parameter ranges where there are no equilibrium solutions in the absence of the harmonic forcing. In this case the constant ambient pressure P_∞ is so low that the tension $p_V - P_\infty$ exceeds the so called Blake's critical threshold. In contrast to the previous papers the evidence of a stabilization process of harmonic excitation is demonstrated for large amplitude oscillations.

The bubble model was the classical Rayleigh–Plesset equation which is a second order highly nonlinear ordinary differential equation. In contrast to the earlier papers there were no restrictions for the nonlinearity of the model, such as linearization or other low order approximations. The application of the sophisticated boundary value problem solver together with the pseudo-arclength continuation technique, all the stable period 1 solutions could be determined in the pressure amplitude – excitation frequency parameter space called phase diagram at different values of the ambient pressures. A large domain of stable period 1 solutions was found beyond Blake's critical threshold proving the existence of the stabilization mechanism.

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